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Evolution of public–private partnership models in American toll road development: Learning based on public institutions' risk management☆

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Abstract

Over the past few decades public-private partnership (PPP) models adopted by governments for infrastructure development throughout the world have evolved continuously. This article develops a dynamic framework which argues that PPP models evolve when some of the critical success factors (CSFs) for PPP are changed/improved over time based on project sponsors' risk management. The framework consists of four elements: CSFs for PPP, rising risks due to poorly addressed CSFs, the corresponding risk management to change/improve the CSFs, and consequently changed PPP models. Here, CSFs for PPP contain three aspects: external environment, internal project characteristics, and partnership-related factors. The framework is empirically explored with a multiple-case analysis of six toll roads developed in the United States since the late 1980s. The results demonstrate a two-phase evolution of PPP models in the studied context, confirm the theoretical framework, and find that public institutions' risk management can effectively explain the PPP evolution.
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1. Introduction

In the past few decades, governments at all levels across the world have been actively exploring a variety of public-private partnership (PPP) strategies for the delivery of public infrastructure and services, driven by growing public demand for infrastructure and services coupled with public sector financial shortfalls and the need for greater efficiencies in project delivery and operations. PPP refers to a contractual agreement formed between public and private sector partners, by which the private partners play a greater than traditional role to design, construct, finance, operate, maintain, or renovate a facility or system (Bovaird, 2004). Notably, PPP is different from privatization. In PPP, the public sector would sign a service contract with the private sector and retain a substantive role in project development. In the case of privatization, public assets are sold to private operators and subsequent government involvement is minimal unless regulation of the post-privatized entity is necessary. There are many specific models of PPP, defined by stages of project delivery and the involvement of private expertise, finance, or other resources. Most common ones include Design-Build-Finance (DBF), Build-Operate-Transfer (BOT), Design-Build-Finance- Operate-Maintenance (DBFOM), and so on (Zhao, 2011).

Notably, PPP models adopted in a field tend to change over time. For example, the state of Victoria, Australia utilized various PPP models to deliver public hospital care during the period 1997–2004. A Build-Own-Operate (BOO) model was adopted for earlier projects with private partners assuming full responsibility for service provision (including financing and ownership), while later projects shifted to a Build-Own-Operate-Transfer (BOOT) model with private partners providing only ancillary services and not keeping ownership of the assets (English, 2005). This is not an isolated incident. Evolution of PPP models has

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been documented in infrastructure-based social services delivery in Europe (Almqvist and Hogberg, 2005), local government service provision in the United States (US) (Hefetz and Warner, 2004), and a variety of other contexts throughout the world (Miranda and Lerner, 1995; Warner and Hebdon, 2001).

Literature has studied why PPP evolution takes place from different perspectives. Scholars studying local government service provision tend to agree that it is mainly for the purpose of seeking more efficient, high quality, and innovative service delivery in a changing context (Hefetz and Warner, 2012; Lavery, 1999; Sclar, 2001; Warner and Hebdon, 2001). While economists argue that incomplete contracts of PPP projects might have accounted for PPP changes. Because of project complexity and uncertainty, the long term of partnership, and a wide variety of risks that might impact partnership (De Bettignies and Ross, 2009), infrastructure PPP projects are likely to have incomplete contracts, which must constantly be revised and/or renegotiated as time goes on, leading to changes of PPP models (Klein et al., 1978; Williamson, 1985). Meanwhile, private developers of infrastructure which own the production technology have strong bargaining power relative to government and might go cherry-picking: they fully involve themselves in the most profitable projects while put less effort into other projects (Koppenjan and Enserink, 2009; Van Ham and Koppenjan, 2001), resulting in the use of different PPP models.

Less well understood is how the evolution of PPP models proceeds and what the nature of the evolution process is. These questions are of particular importance to the US. The nation is experiencing severe funding shortages in infrastructure. The American Society of Civil Engineers (ASCE, 2013) estimates the investment needed for renewing America's infrastructure by 2020 to be $3.6 trillion, yet no funding source is projected to have the capacity to provide sufficient funds. With the encouragement of the federal government, some states have been experimenting with a variety of PPP models as alternatives to traditional project financing and delivery. A better understanding of PPP evolution will not only increase our knowledge and awareness of the dynamic nature of PPP adoption, but also provide valuable information to public administrators seeking more effective and efficient service delivery. This study focuses on these questions for public infrastructure services in general and toll roads in the US in particular. To the author's best knowledge, this is the first empirical analysis to study the process of PPP evolution in infrastructure development.

This paper is organized as follows. The next section reviews literature on government restructuring and critical success factors (CSFs) for PPP. The author then develops a dynamic framework of PPP evolution and empirically explores the framework with a comparative case analysis of six toll road projects developed in the US since the late 1980s. The paper concludes with a discussion of the implications of the results for PPP development and recommendations for future studies.

2. Literature review

2.1. Government restructuring

Governments adopt a wide array of delivery alternatives through using both private and public sector mechanisms to provide public infrastructure and services. The decision of changing the form of service delivery, also called government restructuring, is complex, which balances efficiency goals with public values and attends to the competitiveness of both public and private markets for government services (Hefetz and Warner, 2012; Warner and Hebdon, 2001). Literature has extensively studied the factors determining the decision of changes in service delivery, including efficiency, incomplete contract, monitoring difficulty, service quality, fiscal stress, wealth, and leadership capacity (Hebdon and Jalette, 2008; Savas, 2000; Sclar, 2001; Warner and Hebdon, 2001; Williamson, 1985).

However, empirical knowledge about how the changes in the form of service delivery take place has been fairly limited. For example, Local Government New Zealand (LGNZ, 2003) reported that New Zealand, an early and radical innovator promoting extensive privatization, enacted a new local government law in 2002 to reverse privatization and reassert a government role (Warner, 2008). Martin (2002) also found that the United Kingdom, another early leader in practicing privatization, experienced a similar process of reversing privatization through the political changes brought by the election of Tony Blair in 1997, including a "best value framework" campaign. In addition, learning literature suggests that policy changes may take place through institutional internal learning arrangements, which consist of certain institutional structures, procedures, customs, rules and incentives that support the learning process (Busenberg, 2001; Huber, 1991; Popper and Lipshitz, 1998). Despite the increasing interest in government restructuring, a coherent conceptual framework has not been developed to systematically examine the evolution process of service delivery.

2.2. Critical success factors (CSFs) for PPP

This study intends to explain PPP evolution. To do so, we must first identify critical success factors (CSFs) for PPP, which refer to the key areas of activity where favorable results are absolutely necessary for management success of PPP projects (Hwang et al., 2013) and can influence the selection of a PPP model for a project in practice. CSFs for PPP have been extensively studied from different perspectives, such as for winning BOT contracts (Gupta and Narasimham, 1998; Tiong, 1996; Tiong and Alum, 1997) and for the purpose of formulating and building effective partnerships (Chan et al., 2010b; Chou et al., 2012; Ng et al., 2012; Zhang, 2005).

Prominent in this literature are studies that consider a full range of external and internal factors which might facilitate or impede PPP implementation. Zhang (2005) develops a CSFs package for improving PPP procurement protocol to achieve a win-win relationship. The package contains five groups of CSFs, namely: (1) favorable investment environment, (2) economic viability, (3) reliable concessionaire consortium with strong technical strength, (4) sound financial package, and (5) appropriate risk allocation via reliable contractual arrangements. Each of the five includes a number of CSFs. In a study on international construction joint ventures, Ozorhon et al. (2007) suggest that the external environment under which the ventures operate and internal project-related factors have a positive effect

on ventures' performance. Here, the environment refers to factors such as political stability, macroeconomic conditions, strength of the legal system, and the relations between the venture partners and the host government. Internal project-related factors include availability of resources, technical complexity of the project, completeness of the design, completeness of the contract documents, project requirements in terms of quality, environment, health and safety, and so on.

3. A dynamic framework of PPP evolution

This study develops a dynamic framework of PPP evolution, which argues that PPP models evolve when some of the CSFs are changed or improved over time based on project sponsors' risk management, as illustrated in Fig. 1. The framework is considered dynamic because it proposes that four elements: the CSFs of PPP development (Element 1), rising risks due to poorly addressed CSFs (Element 2), the corresponding risk management to change/improve the CSFs for future projects (Element 3), and consequently changed PPP models (Element 4), together constitute the operation and feedback loops to push the PPP evolution process. Here, Element 1 contains three aspects of CSFs for PPP, including external environment, internal project characteristics, and partnership-related factors. Notably, when this study talks of success or failure, it relates primarily to the purpose of formulating and building viable, stable, and effective PPP.

Among various methods of CSFs classification, this study adopts the external vs. internal factors model suggested by Ozorhon et al. (2007), synthesizes relevant concepts developed within the CSFs literature, and then modifies the model to serve the research purpose of exploring PPP evolution. Besides external environment and internal project characteristics, the author complements the model by considering a third perspective, partnership-related factors. It is because empirical research has found that some factors related to the way public and private partners collaborate are of particular importance to the success of PPP, such as PPP-related government capacity (Hwang et al., 2013; Kort and Klijn, 2011), the selection of private partners (Palaneeswaran and Kumaraswamy, 2000; Zhang, 2005), and the division of responsibilities and risks among partners (Chan et al., 2010b; Ozorhon et al., 2007). As a result, this study considers three aspects of CSFs for PPP development (Element 1): external environment, internal project characteristics, and partnership-related factors.

Based on the CSFs classification, the study proposes that project sponsors' risk management plays a crucial role in driving PPP evolution through its impact on changing/improving the CSFs. According to both bounded rationality (Ostrom, 2007) and incomplete contract theories (De Bettignies and Ross, 2009), it is unrealistic to expect a new policy or program to work perfectly as expected without the need for modification or changes. As an infrastructure project proceeds, certain risks may arise when some CSFs are not well addressed (Element 2). For example, lack of political backup and public support might bring political risk. When these risks are major, project sponsors would adopt corresponding risk management strategies as a response to better address these CSFs in the future (Element 3), leading to a change of the PPP model for future projects (Element 4). This evolution cycle goes on as long as adoption of a PPP model creates major risks. Notably, not all risks are manageable; for example financial risk caused by the worldwide financial crisis is probably beyond the control of any project sponsor. Hence, the PPP evolution cycle may end in one of the following situations: (1) the rising risks are beyond project sponsors' capability to manage; (2) the costs of additional risk management are viewed as too great; or (3) a PPP model is judged good enough to serve current needs.

3.1. CSFs: external environment

Literature suggests that four environmental factors are particularly important to the development of effective PPP: overall infrastructure demand, financial situation, political ideology, and relevant legislation (Gómez- Ibáñez and Meyer, 1993; Wang and

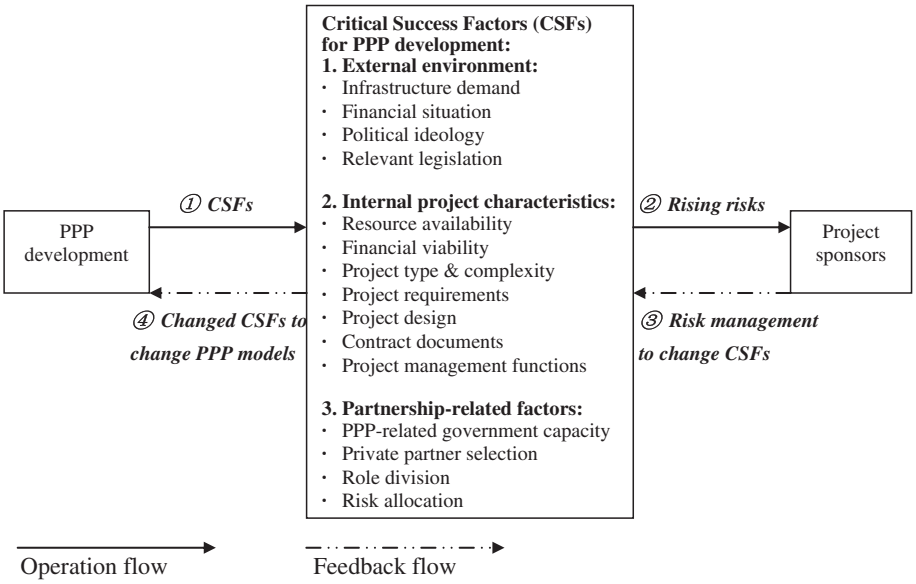


Fig. 1. Evolution of PPP models based on project sponsors' risk management.

Zhao, 2014). Infrastructure demand matters as large aggregate demand brings good chances of investment returns, and hence attracts the private sector to participate (Hammami et al., 2006). Another primary external factor for PPP adoption is economic -- to seek new ways to meet growing demand in the context of fiscal constraints (Vining et al., 2005) and deliver improved public services at lower cost (Siemiatycki and Farooqi, 2012).

Thirdly, politics plays a significant role in the PPP debates (Chan et al., 2010a). On one hand, worldwide political campaigns associated with the elections of Ronald Reagan in the US and Margaret Thatcher in Great Britain changed the preferences of citizens and politicians about the proper role of government and produced efforts to reduce the size of government and increase the private sector's role in public service delivery that continue today (Rainey, 1990; Wettenhall, 2005). On the other hand, private involvement in public affairs may arouse some political concerns. Public sector employees and their unions may oppose PPP due to the implications for public jobs (Wang and Zhao, 2014) and citizens may also be concerned about the loss of public control over key public infrastructure (Baxandall et al., 2009). The fourth factor is PPP legislation, which shapes the legal and regulatory environment within the jurisdiction for partnership formation, operation, and sustainability. Strong and effective legislation tends to decrease the uncertainties of government regulation and increase the chances of success for PPP projects (Hammami et al., 2006).

3.2. CSFs: internal project characteristics

Project characteristics are widely considered important factors affecting PPP success. Important ones include availability of financial and other resources, project financial viability, project type and the associated technical complexity, project requirements regarding government permissions, quality, environment, health, and safety, completeness of project design, completeness of contract documents, and effectiveness of project management functions such as planning, coordination, monitoring, and controlling (Bolton et al., 1994; Koppenjan, 2005; Ozorhon et al., 2007). The strong impact of project characteristics on PPP effectiveness and performance reinforces the need of project sponsors handling project-related risks with extra care through effective risk management (Mohamed, 2000).

3.3. CSFs: partnership-related factors

Building PPP is an interactive process which requires close and intensive communication, negotiation and collaboration between the public and private sectors. Literature suggests four important partnership-related factors: PPP-related government capacity, private partner selection, role division among partners, and risk allocation among partners (Koppenjan, 2005; Yang et al., 2013). First, PPP-related government capacity refers to the expertise, knowledge, decision-making mechanisms, administrative systems and credibility that government agencies possess with regard to negotiating, operating, and supervising PPP projects (Yang et al., 2013). Such capacity is a prerequisite for public agencies to successfully initiate, contract, and manage PPP projects (Brown and Potoski, 2003a,b). Secondly, selecting the right private partner is a crucial step towards building successful partnership (Hwang et al., 2013). The literature suggests two broad categories of benefits that government seeks from collaborative partners: it may seek a resource from a partner that it needs for its activities but does not possess (resource exchange), or it may seek associated advantages from a well-respected or well-connected partner (organizational legitimacy) (Dyer and Singh, 1998). Meanwhile, mutual interdependence among partners implies vulnerability to the behavior of one's partner (Graddy and Chen, 2006). To mitigate the vulnerability, the public agency must identify a partner it views as trustworthy. Here trust means positive expectations of both partner intentions and its competence (Das and Teng, 2001). Working with such a partner can reduce transaction costs associated with partnership and enhance the chances of success for PPP.

Furthermore, division of roles among public and private parties is also critical as empirical evidence shows that a clear delineation of roles and responsibilities would reduce transaction costs and uncertainty (Graddy, 2009). Generally speaking, the public sector in PPP plays a directing role, organizing project activities and setting game rules. The private sector usually plays a role of doer, taking care of technical specifications, physical construction and daily management (Kort and Klijn, 2011; Van Ham and Koppenjan, 2001). Of course, these are only general rules and specific arrangements in a project must go through rounds of negotiation before being formalized in the form of a signed contract.

The fourth factor is risk analysis and allocation (Ozorhon et al., 2007; Zhang, 2005). Risk exists "when a decision is expressed in terms of a range of possible outcomes and when known probabilities can be attached to the outcomes" (Smith et al., 2006). Although risk might refer to the chance of either bad or good consequences, in the context of infrastructure development, the author focuses on the risk of bad consequences that could prohibit smooth project development or jeopardize project viability. Literature suggests infrastructure PPP projects face at least eight major risks: political risk, legislative risk, administrative/regulatory risk, financial risk, revenue risk, design and construction risk, operating and maintenance risk, and contract risk (Baldry, 1998; Chou et al., 2012; Grimsey and Lewis, 2002; Zhao, 2011). A general rule is to assign project risks to the parties best able to manage them (Bing et al., 2005; Van Ham and Koppenjan, 2001), because each party to a partnership has a level of risk tolerance and a capacity to manage certain risks. This method is considered an effective way to reduce project costs and improve the potential for success. It, however, is not always an easy determination so risk allocation remains a complex issue in PPP implementation.

3.4. Rising risks, risk management, and the consequent PPP model changes

Infrastructure development is an expensive and complicated endeavor and project risks might emerge when some of the CSFs are not well addressed. As discussed in Section 3.3, the study pays particular attention to eight types of risks (political, legislative, administrative/regulatory, financial, revenue, design

and construction, operating and maintenance, and contract risks). To tackle these risks, an effective risk management process (RMP) should embrace both the traditional model proposed by the risk management literature (Chapman and Ward, 2003; Tummala and Mak, 2001) and the recent AS/NZS ISO 31000:2009 risk management standards (Leitch, 2010; Purdy, 2010),[1] including the following steps: (1) before beginning the process of assessing risk, project sponsors must establish the detailed context, which sets the scope and risk criteria for the process; (2) the RMP then proceeds to risk identification, risk measurement, and risk assessment, during which project sponsors identify potential risks, enumerate the associated consequences, and assess their severities and probabilities of occurrence; (3) the risk evaluation and control phase involves the solution generation stage consisting of ranking the identified risks, developing risk control action plans, and selecting the best course of action; and risk control has five major approaches, namely accept, reduce, avoid, spread and transfer (Tummala and Mak, 2001); and (4) meanwhile two key functions should happen continually throughout the RMP, namely communication and consultation (including both internal and appropriate external stakeholders), and monitoring and review (reviewing the progress of all risk management activities).

Risk management is expected to change the CSFs which caused the risks in the first place, in turn leading to changed PPP models for future projects. It is because changing external environment, different project characteristics, and modified/improved partnership between public and private partners would all affect the decision of a project sponsor about the most suitable PPP model for an infrastructure project.

4. Empirical analysis

This research empirically explores the dynamic framework of PPP evolution with an analysis of PPP toll road development of state transportation agencies in the US since the late 1980s. Guided by the dynamic framework, attention was paid to the CSFs of these PPP toll roads, major risks state transportation agencies encountered during project development due to poorly addressed CSFs, the corresponding risk management strategies, and the consequent changes to the CSFs. Analysis focused on a causal chain among these factors, especially how the factors contribute to PPP evolution through a dynamic loop. Notably, the study is of an exploratory character and its purpose is to increase our knowledge and awareness of PPP evolution and the effects of risk management on this process. The proposed dynamic framework may form the basis for subsequent omnifaceted evaluation of the course and outcomes of PPP evolution and further offer indications for the improvement of the quality and effectiveness of evolved PPP models.

PPP toll roads are relatively new to the US, which didn't emerge until the late 1980s. The construction of the Interstate Highway System, the world's largest freeway system, essentially came to an end around the mid-1980s. Since then state governments started to experience financial shortages in transportation field due to rapidly increasing travel demand, inflation, improved vehicle fuel efficiency and increased construction costs (TRIP, 2014; Wachs, 2006; Wang and Zhao, 2014). With the passage of the 1991 Intermodal Surface Transportation Efficiency Act (ISTEA), the first federal legislation to authorize tolling and private financing for transportation projects (USGAO, 2004), some states have turned to private finance and experimented with a variety of public-private collaboration arrangements for roads. It is this experimentation that allows the author to explore the evolution of PPP models in the field.

The research population consists of toll road projects involving private finance that were initiated after the late 1980s and have successfully gone through the formation process and reached public-private contracts as of December 2012. In the US this population is limited in scope, 29, to be exact, based on the data the author collected. For the empirical analysis, the author relies on a comparative case methodology by selecting a number of suitable cases from the research population. This approach is necessitated here by the limited number of PPP projects and the multiplicity of factors believed to impact the process of PPP evolution.

The author used the following criteria to select cases. First, the selected cases should cover all innovative public-private arrangements adopted by state transportation agencies since the late 1980s so that we can observe how PPP models have evolved over time. Secondly, these projects should have passed the initial planning, financing and design stages, and reached public-private contracts by the time this study is being conducted. Since only 13 out of the 29 PPP roads had been completed and opened to traffic by December 2012, to avoid loss of crucial information, the author cannot exclude all uncompleted projects and has to be satisfied with projects that have passed the planning, financing, design and contracting stages, where most known project risks take place. Third, in order to justify generalizations about state of the art in the field, the studied projects should be from states most active in PPP roads with project initiation time across the past few decades. Ultimately, the following six projects were selected for analysis, in order of their initiation time:

- *Dulles Greenway:* initiated in 1988, a fourteen-mile toll road, serving the anticipated traffic growth near Dulles Airport and Loudon County, Virginia. It was the first privately financed toll road in the nation, developed through an unsolicited proposal by a local consortium Toll Road Investors Partnership II (TRIP II) with full private funding of $350 million, under a Design-Build-Finance-Operate contract with Virginia Department of Transportation (VDOT). The highway opened in 1995 and TRIP II would own and operate it for 42.5 years. Due to low traffic flow, the road encountered severe revenue shortfalls and went into default on its loans in 1996. Consequently, the state extended the concession period an additional 20 years in 2001 and the road was sold to Macquarie in 2005.
- *CA 91 Express Lanes:* initiated in 1989, a ten-mile long toll highway, within the median of pre-existing SR 91, serving

[1] AS/NZS ISO 31000:2009 refers to a family of standards relating to risk management, codified by the International Organization for Standardization. I thank a reviewer for suggesting including AS/NZS ISO 31000:2009 in the RMP.

southern California. It was developed by a domestic firm California Private Transportation Company (CPTC) with full private funding of $130 million under a PPP agreement with California Department of Transportation (Caltrans) and local government. The project opened to traffic in 1995 and CPTC retained the franchise for the road for 35 years. Notably, a non-compete clause in the contract severely jeopardized the public interest by preventing the state from making improvements to the competing highways, leading to public takeover of the road in 2003 (USGAO, 2004).

- *South Bay Expressway:* initiated in 1991, a 12.5-mile toll highway located in San Diego, California. It was developed by a domestic company California Transportation Ventures (CTV) under a PPP agreement with Caltrans and local government. Notably, the project development agreement was reached in 1991 but the expressway didn't open to traffic until 2007 due to an extremely lengthy and expensive environmental clearance process, which cost the developer nine years and $50 million (AECOM, 2007a). The project cost $722 million. Upon completion CTV retained the franchise for the road for 35 years. Since its opening, the road had encountered severe revenue shortfalls due to the recent economic downturn in the US, which lead to public takeover of the road in 2011.
- *Pocahontas Parkway:* initiated in 1995, an 8.8-mile toll facility serving Chesterfield and Henrico counties, Virginia. The project was developed under a Design-Build-Finance-Operate contract between the state and a private non-profit group Pocahontas Parkway Association (PPA), a joint venture of two international infrastructure companies (Fluor Daniel and Morrison Knudsen). The $381 million project was financed by innovative financial tools including non-profit bonds issued by PPA and SIBs.[2] The road opened to traffic in 2002 and the developer would retain the franchise for the road for 30 years. Since its opening, the road has had severe revenue shortfalls, leading to a sale to Transurban.
- *I-394 MnPASS:* initiated in 2004, an eleven-mile toll road converting previous High Occupancy Vehicle (HOV) lanes on I-394 to Highway Occupancy Toll (HOT) lanes, located in Minneapolis, Minnesota. The $12.5 million project was developed under a PPP contract between Minnesota Department of Transportation (MNDOT) and Wilbur Smith Associates (an international infrastructure firm) and opened to traffic in 2005. MNDOT made much effort to obtain public support and political backup for the project. Since opening, financial performance of the road has been a big disappointment, partly because the HOT lanes have significantly improved traffic flows on the free lanes so that drivers lack incentives to use the tolled lanes.
- *The New LBJ Managed Lanes:* initiated in 2009, a thirteen-mile project to expand I-635 to fourteen lanes from eight, located in Dallas, Texas. The state executed a Comprehensive Development Agreements contract with a consortium led by a foreign toll road company Cintra to enter a 52-year toll concession to develop and operate the project. The $2.7 billion project received both public and private funding and used innovative financial tools including private activity bonds and TIFIA.[3] The road is currently under construction for completion by 2016. The fact that the consortium is led by a foreign company has caused severe political backlash.

Information of these projects was collected from multiple sources to ensure accuracy, including academic papers, government reports, relevant program/project documentations from federal and state transportation agencies and project official websites, archival news reports, and interviews. The author identified three types of project stakeholders for interviews: state transportation agencies, private developers, and transportation consultants. Archival written materials on the selected cases offered some clues as to who seemed to have played a key role in these cases. The author specifically sought to interview people with much first-hand experience of negotiating on contract terms and risk management strategies. As the interviews proceeded, interviewees were asked who else should be interviewed, which significantly widened the access to key players involved in these projects. As a result, the author interviewed seven persons: two policy makers from state transportation agencies, two managers from private developers, and three transportation consultants. The interviews were semi-structured with a set of questions as a guide and various follow-up issues and questions were explored as the interviews progressed. Appendix 1 serves as an example of interview questionnaire which was particularly designed for transportation consultants. Interview time per person ranged from one to one and half hours. Detailed notes were taken in the course of the interviews, and transcribed in the hours following the interviews.

The data collection was guided by the dynamic framework, covering the three aspects of the CSFs for these PPP projects, major risks they encountered, the corresponding risk management strategies, and lessons about how to better address the CSFs for future projects. Specifically, the information collected for each selected project includes: (1) project external environment (politics, financial situation, and legislation), (2) project internal characteristics (facility details, project goals, project complexity, concession term, and contract clauses regarding environmental review processes, land acquisition, and termination for convenience), (3) partnership-related information (partner selection, the roles each partner plays, and financing arrangements); (4) major risks encountered and the causes; and (5) the corresponding risk management measures for future PPP. Analysis of the qualitative data was focused on the causal chain among these factors which eventually forms the dynamic loop of PPP evolution. For the purpose of examining the role of risk management in affecting the evolution of PPP models through its impact on changing/

[2] State Infrastructure Banks (SIBs), together with the Transportation Infrastructure Finance and Innovation Act of 1998 (TIFIA) and private activity bonds, are three important programs authorized by the United States Federal Government to provide financial assistance to support certain transportation projects. Please see "Project Finance Primer 2010" by FHWA http://www.fhwa.dot.gov/ipd/finance/resources/general/default.aspx for details.

[3] Please refer to footnote 2.

improving the CSFs, Table 1 highlights the encountered risks and well managed risks (based on learning from previous projects' risks) in each project, along with the CSFs which are subject to project sponsors' risk management.

5. Findings and discussions

The findings of the comparative case analysis are presented in two parts. First, evidence is found of a two-phase evolution of PPP models in this infrastructure context. Second, when exploring the reasons for this pattern, the author finds that project risks arise when some CSFs for PPP are not well addressed, and that public agencies sponsoring the PPP projects learn the lessons and take corresponding risk management measures to better manage these CSFs in the future, consequently leading to a change of the PPP model for future projects.

5.1. Observed evolution of PPP models

There has been an evolution of PPP models in American toll road development in the past few decades: a private development model emerged around the late 1980s, under which the private sector takes full responsibility for project development while the public sector plays a minimized, supporting role; and a collaborative partnership model was first adopted around the mid-1990s and boomed in the new millennium, under which public and private entities assume shared but differentiated responsibility in almost all aspects of project development.

A private development model first emerged in the US around the late 1980s when some states first began to consider private financing and tolling due to severe shortages in funding for transportation services. This model involves maximum private participation in that responsibilities for project development and the associated risks are completely transferred to the private sector. A private entity is authorized to fully finance, develop, operate and maintain a toll road, and meanwhile given sole entitlement to project-generated revenues (USGAO, 2004; Vining et al., 2005). The public sector, although sometimes retaining project ownership and acting as the project sponsor, plays primarily a regulatory role in guiding administrative processes and overseeing project development. This model has been adopted in some real estate development cases (Samuel, 1998) and also early toll road concessions (Bovaird, 2004; FHWA, 2004). Thus far, the nation has built ten projects under this model. In this study, Dulles Greenway, CA 91 Express Lanes, and South Bay Expressway are the pilot projects developed under this model.

A collaborative partnership model gradually emerged around the mid-1990s and boomed in the new millennium. In contrast to the previous private development model, under this model public and private entities assume shared but differentiated responsibilities and risks in most aspects of project development. Specifically, both sectors make financial contributions to project capital and share the entitlement to project revenues, in addition to their collaboration in other aspects such as design, construction, operation and maintenance. Hence this model involves much deeper and more complex collaboration among public and private parties. Recent toll concessions have mostly adopted this model, under which around twenty projects have been developed, including Pocahontas Parkway, I-394 MnPASS, and the New LBJ Managed Lanes studied in this research.

5.2. How the evolution proceeded: learning based on public institutions' risk management

The author finds that public institutions' risk management can effectively explain the PPP evolution. As seen in Table 1, over time public institutions have (1) built a more supportive environment for PPP adoption by addressing political and legislative risks, (2) better managed project characteristics by handling project financial, revenue, administrative/regulatory, and contract risks, and (3) deepened the collaboration with the private developers via developing government capacity for PPP, selecting suitable private partners, and making effective risk transfer and responsibility sharing, which together can well explain the transit from the private development model to the collaborative partnership model.

5.2.1. Building a more supportive environment

Environmental factors considered in this study are state level factors, including transportation demand, financial situation, political ideology, and the related PPP legislation. Increasing transportation demand and severe financial constraints are the two most important external factors motivating the utilization of PPP in American toll road development. Their effects have been consistent and ever-increasing in the past few decades.

In contrast, political attitudes towards PPP and the related state legislation have gone through more complex learning processes and the associated political and legislative risks are noteworthy. Interviews revealed that early PPP projects built under the private development model all incurred large political costs with almost every political leader who promoted privatization suffering negative political consequences. Public concerns appeared in many states with particularly strong opposition from highway construction and maintenance contractors, government engineering consultants, public employees, and the municipal bond industry. Moreover, early PPP projects often lacked effective authorization legislation. A typical example is the 1989 California Assembly Bill 680, under which six out of eight authorized projects were never able to take off due to lack of finance, financial viability or community support, and CA 91 Express Lanes and South Bay Expressway were the only ones eventually built (Gómez- Ibáñez and Meyer, 1993). Over time, this bill earned a reputation as a PPP legislative model to avoid and was repealed by the California state legislature in 2002.

With these lessons learned, state transportation agencies took a series of actions to manage political and legislative risks for later projects. Firstly, public agencies made more efforts to gain political backup and public support. For example, Minnesota Department of Transportation (MNDOT) proposed the I-394 MnPASS project in 1996 for the first time but failed because of political sensitivity of tolling and lack of strong political backup. When MNDOT made its second effort for

Table 1
Three aspects of critical success factors, the encountered risks and well managed risks of the six toll road projects.

PPP model	Project (initiation)	External environment	Internal project characteristics	Partnership-related factors	Encountered risks and well managed risks based on learning from previous PPP
Model 1: Private development model	Dulles Greenway (1988)	State had a large backlog of needed projects and financial shortage; developer convinced state to pass the legislation.	Full private financing; uncertain financial viability as a new development road; needed to acquire land and environmental permits.	Insufficient government capacity; a local developer submitted an unsolicited proposal, assumed full project responsibility & risk.	*Encountered risks:* revenue shortfalls causing the state to extend the concession period another 20 years.
	CA 91 Express Lanes (1990)	State had a large backlog of needed projects and financial shortages, passed Assembly Bill 680 to authorize PPP; political concerns with privatization.	Full private financing; good financial viability as a congestion relief road; obtained land from state, purchased environmental studies from local government; a non-complete clause in contract.	Insufficient government capacity; a domestic firm initiated project in response to a general call of state, assumed full project responsibility and risk; government supported land and environmental work.	*Encountered risks:* political concerns with privatization; ineffective legislation; contract risk with the non-compete clause, which severely jeopardized the public interest, leading to a public buyback.
	South Bay Expressway (1991)	State had a large backlog of needed projects and financial shortages, passed Assembly Bill 680 to authorize PPP; political concerns with privatization.	No public funds allowed in original contract; uncertain financial viability as a new development road; needed extensive environmental and land work; a non-complete clause in original contract.	Insufficient government capacity; a domestic firm initiated project in response to a general call of state, assumed full project responsibility and risk; government helped with financing later.	*Encountered risks:* political concerns with privatization; ineffective legislation; severe revenue shortfalls lead to public takeover; administrative/regulatory risk with costly, lengthy environmental clearance; ineffective risk transfer to developer.
Model 2: Collaborative partnership model	Pocahontas Parkway (1995)	State had a large backlog of needed projects and financial shortages, passed the Public Private Transportation Act in 1995.	Funded with non-profit bonds and SIBs; predicted with financial feasibility; needed little environmental and land work due to prior government studies; contract had limited non-compete and termination clauses.	Improved government capacity; a private non-profit venture with international background submitted an unsolicited proposal, shared project responsibility and risk with state.	*Well managed risks:* effective legislation; innovative financial resources; reduced administrative risk with little land and environmental work needed; improved contract terms including a termination clause & limited non-compete protection. *Encountered risks:* the non-profit model didn't work well; revenue shortfalls.
	I-394 MnPASS (2004)	State worked hard to mitigate political risk and gain public support, passed the High Occupancy Toll Lane Legislation in 2003.	Both public and private financing; good financial viability with conversion of HOV to HOT lanes; no environmental or land work needed.	Improved government capacity; state selected an international professional toll road firm via a Request- for-Proposal (RFPs) process, shared project responsibility and risk with the firm.	*Well managed risks:* reduced political risk by public campaign; no administrative /regulatory risk related to environmental or land requirements. *Encountered risks:* revenue shortfalls due to improved traffic flow on free lanes.
	The New LBJ Managed Lanes (2009)	State had severe traffic congestion, needed large capital; PPP law was enacted in 2003, amended in 2005; public concerns with foreign takeover of highways.	Public and private financing, TIFIA, private activity bonds; good financial viability with existing congestion; needed little land or environmental work; a termination clause and no non-compete restriction.	Improved government capacity; state selected an international professional toll road firm via a RFPs process, shared project responsibility and risk with it.	*Well managed risks:* effective legislation; innovative financial resources; good financial viability; little administrative/ regulatory risk related to environmental or land work; improved contract with a termination clause and no non-compete restriction; effective risk transfer. *Encountered risks:* political concerns of foreign takeover.

the project in the early 2000s, it started with asking for public support and political backup through actions including conducting an opinion survey and holding several local public hearings. It also actively involved state political leaders this time including State Governor and state transport commissioner (Samuel, 2005). The obtained support helped make the project a smooth, successful PPP. Secondly, states have become more sophisticated in making effective legislation to guide PPP development. For instance, the 2003 Texas PPP legislation (HB 3588), which was strengthened by amendments in 2005, attracted enormous private capital investments for the state's highway development, including the LBJ Managed Lanes studied in the research. The 2003 Texas legislation, along with the 1995 Virginia Public-Private Transportation Act under which Pocahontas Parkway was authorized, has been cited repeatedly as model by other states enacting similar legislation.

5.2.2. Better managed project characteristics

Four project characteristics, including resource availability, financial viability, project complexity and requirements, and contract terms, are found to be particularly important to PPP development in this infrastructure context. The associated project risks, namely financial, revenue, administrative/regulatory, and contract risks, arose under the earlier private development model. Risk management strategies to better address these project characteristics have contributed to the evolution of PPP models.

The first was financial risk. Resource availability was always a problem for early projects under the private development model because the private sector was left alone to seek capital and all other needed resources, which constrained the scale and complexity of these projects. Secondly, there was revenue risk. Financial viability of early projects was often uncertain because most of them were new development roads with a purpose of serving anticipated traffic growth in the region (such as Dulles Greenway and South Bay Expressway) and their toll revenues were highly subject to events such as slowdown in local economic development and changes to regional road systems. The third was administrative/regulatory risk. New development roads going through undeveloped areas often require complex project design and need to meet a series of technical and administrative requirements such as environmental clearance and land permission. Evidence shows that these requirements could be very costly, even devastating in some cases. For example, the South Bay Expressway was severely challenged in the environmental review processes. It took the developer $50 million and nine years to receive the final environmental clearance (USGAO, 2004). Lastly, there was contract risk as some contract terms of the early PPP projects were found not serving the public interest. The non-compete clause in the contract of CA 91 Express Lanes, for example, prevented the state from making improvements to the competing SR 91 freeway, which the state claimed was necessary for driver's safety. The local government had to pay an inflated price to buy the facility back to solve the deadlock (Vining et al., 2005).

These project-related risks have received much attention from public agencies, which have taken effective risk management strategies, such as risk avoidance, reduction, spread and transfer, to better address these project characteristics under the later collaborative partnership model. Firstly, financial risk was reduced. Federal government has authorized a set of innovative financial tools such as SIBs, TIFIA, and private activity bonds to provide later projects with better access to resources and meanwhile the public sector, especially the federal government, has increased its grant contribution to these projects. With the improved resources, these projects were able to provide larger-scale, better-designed road facilities with longer concession terms. Secondly, later PPP projects have improved financial viability. Public agencies have learned that the key to ensure a toll project's viability is having real traffic congestion, and therefore they gave up developing new development roads and chose to build roads with existing congestion (as seen in I-394 MnPass and the New LBJ Managed Lanes). Thirdly, to reduce project technical complexity and administrative requirements for later projects, states often deliberately chose to build roads that had been long planned and studied so that the projects wouldn't encounter serious problems with regional planning requirements and rules. Meanwhile states usually completed the environmental and land work before the award of partnership contracts (Pocahontas Parkway), or worked closely with developers to minimize the related problems (The New LBJ Managed Lanes). Lastly, the public sector has managed to add new terms in later partnership contracts to better protect the public interest, for example a termination for convenience clause. This clause is a lesson of the non-compete clause of the CA 91 Express Lanes, which ensures government the rights to take over a privately developed facility when it becomes necessary without breaching a partnership contract.

5.2.3. Deepened collaboration between the public and private sectors

This study examines four partnership-related factors that are critical to the success of PPP development: PPP-related government capacity, private partner selection, role division, and risk allocation. The analysis shows that these factors have caused certain concerns under the earlier private development model, namely opportunistic behavior of private partners, incompetent partners, and ineffective risk transfer to the private sector. Risk management strategies to better address these factors have contributed significantly to the transit to the collaborative partnership model.

First, there was a concern about opportunistic behavior of private partners. In the early days of PPP, public agencies had insufficient government capacity in developing PPP as they lacked the relevant knowledge, expertise, and institutional mechanisms. State officials didn't feel that they had enough knowledge to identify suitable projects to be built as PPP and therefore encouraged the private sector to initiate PPP projects. However, in more than a few instances, public officials after the fact perceived that private developers took advantage of information asymmetry and their proposals tended to serve the interest of the developers rather than the public. The non-compete clause in the contract of CA 91 Express Lanes serves as an example of this point. Another risk came from incompetent private partners. These early projects were all

developed by domestic firms as no international companies had entered the US market back then. These domestic companies were not professional road developers since toll road development was almost nonexistent in the US before the mid-1980s.[4] They were in fact from all walks of life, such as family business (Dulles Greenway) and communication industry (CA 91 Express Lanes). A problem with these companies was that they often lacked expertise, experience and financial capability to handle tough situations such as severe revenue shortfalls and major engineering obstacles. In fact they often chose to quit a project and end partnership ahead of schedule in such situations, as seen in Dulles Greenway and South Bay Expressway. Thirdly, a complete responsibility and risk transfer to the private sector created serious consequences as the private sector couldn't handle certain tasks well, especially the ones related to public administrative and regulatory processes (AECOM, 2007b). "The environmental marathon" of South Bay Expressway serves as an example of this point.

To tackle these problems, public institutions adopted a series of risk management strategies to reduce, spread, and transfer these risks in later projects. Firstly, with years of experience in toll road partnerships, the public sector has become more knowledgeable about PPP and the improved government capacity can effectively mitigate private opportunism. Some states began to adopt a Request-for-Proposals (RFP) process to initiate a project, by which government designates a desirable project, sends out a RFP to solicit multiple private proposals, and then selects the most suitable developer with the best value proposal (Zhao, 2011). Secondly, the choice of private partners has changed for later PPP projects, which is considered beneficial to partnership stability and effectiveness. Some international professional toll road firms entered the US market around the mid-1990s, such as Flur Daniel, Cintra, Transurban, and Macquaire. These world-class companies are considered more trustworthy than domestic unprofessional developers because of their expertise, experience, and reputation, and therefore were often selected as the developers for the later partnerships (Pocahontas Parkway, I-394 MnPASS, and the New LBJ Managed Lanes). Lastly, the public sector has handled risk transfer and responsibility sharing in a more sophisticated way in the later partnerships. Government played an increasing role in project development and assumed more project risks: (1) states would usually complete environmental and land work before the award of partnership contracts, or work closely with developers to minimize the related problems, and (2) the federal government has become an important financier for the later partnerships by authorizing a set of innovative financial tools including SIBs, TIFIA, and private activity bonds. The greater government involvement in later partnerships has substantially expedited project development and increased project viability.

[4] The 1950s to the mid-1980s was the Interstate Highway Era in the US, during which development of toll roads was explicitly prohibited.

6. Conclusions and recommendations

This study examines the process of PPP evolution in infrastructure development with a comparative case analysis of six toll projects in the US. The analysis first demonstrates a two-phase evolution of PPP models in this context: from a private development model to a collaborative partnership model. Second, the analysis shows that public institutions' risk management can effectively explain the PPP evolution. The research considers three aspects of CSFs for PPP development, consisting of external environment, internal project characteristics, and partnership-related factors. Empirical analysis suggests that, when project risks arise as a result of poorly addressed CSFs, public institutions would take corresponding risk management measures to change the way these CSFs are addressed in the future, consequently leading to a change of the PPP model for future projects.

This study carries some significant practical implications, which can help policy makers in facilitating the formation and implementation of PPP. Given the fact that infrastructure PPP projects are likely to have incomplete contracts and new risks/problems might emerge continually, it is the most critical for public institutions to enhance their capacity in managing PPP and build their institutions and legal systems for an organized and efficient operating environment for successful PPP development. First, government needs to handle political issues with extra care since political concerns towards privatization are common across states and could be intense sometimes. Government may take a series of actions to reduce public concerns and gain political support for PPP projects, such as conducting opinion surveys, establishing a task force with professionals or local notables, holding hearings to seek for suggestions, and involving high-ranking politicians. Second, enactment of effective legislation is also important as it shapes a supporting legal and regulatory environment to guide partnership implementation. The National Conference of State Legislatures (US) has published a toolkit for legislators, which may provide a framework to guide state PPP legislation (Rall et al., 2010). In practice, policy makers may refer to the 2003 Texas PPP legislation (HB 3588), which is acknowledged as the most comprehensive, effective PPP legislation to date, followed by Virginia's 1995 Public-Private Transportation Act. Thirdly, government needs to handle risk transfer more sophisticatedly as complete risk transfer to the private sector may seem attractive at first but hurt the public interest eventually. Government should play a greater role in handling regulatory or administrative requirements of PPP projects, like the environmental clearance and land permission processes in the studied context. Meanwhile government could actively utilize innovative financial tools to supplement private investments and increase project viability.

Another practical implication is that the learning that drives PPP evolution is hardly perfect, also due to contract incompleteness. For example, revenue risk was always present in this study: four out of the five toll roads which have opened to traffic in this study have experienced revenue shortfalls to some extent. It is because low traffic flow (the cause of revenue risk) might be caused by various reasons, such as inaccurate traffic prediction, economic downturn, or improvements to competing freeways.

It is difficult to specify ex ante the best design, construction techniques, or partnership arrangements to fully prevent this risk. As a result, the rising of new problems associated with imperfect learning may motivate another round of evolution.

The author would like to point out some caveats of this study along with suggestions for future research. First, besides risk management, changes of PPP models may be explained by some other factors, such as cherry picking of the private sector or existence of policy entrepreneurship. Private developers may first cherry-pick the most profitable projects through their initiatives and then over time require more public support and subsidies for less favorable projects. Meanwhile, as some literature suggests, policy entrepreneurship may be used to explain policy changes (Klein et al., 2010; Mintrom and Norman, 2009). These two factors, however, were not considered in this study due to lack of relevant information. Second, the proposed framework is to investigate long-term PPP evolution. The fact that only two recognizable PPP models have been utilized in American toll roads in such a short history has limited the robustness of the findings. To tackle these concerns and improve the framework of PPP evolution, future studies could use system dynamics models to embrace more possible evolution drivers and success factors to further explore the course and outcomes of PPP evolution. System dynamics models look at things as a whole system. The central concept is that all the objects and people in a system interact with one another through feedback loops, where a change in one variable affects other variables over time, in turn affecting the original variable (Forrester, 1971; Meadows, 2008; Sterman, 2000). In the studied context, system dynamics models could include other policy change drivers such as cherry picking of private developers and the influence of policy entrepreneurs along with risk management, and be tested with empirical evidence of more PPP models when they are adopted for toll road development.

Lastly, caution should be taken in generalizing this study's results, as the findings are based on US toll roads—a single type of public infrastructure. Different infrastructure types in different jurisdictions vary in the availability of public resources, in levels of private investment interest, or in degrees of public acceptance of PPP. Nevertheless, the theoretical framework and research approach of this article can be applied to examine the adoption of alternative project delivery approaches (including PPP) and their evolution in other types of infrastructure development. Findings from this study can provide a point of comparison for future work of this line, enhancing our understanding of the circumstances under which public decision makers seek more effective and efficient service delivery in public infrastructure as well as other sorts of public service delivery.

Appendix 1. Interview questions for transportation consultants

The following is a list of open-ended questions prepared for multiple situations. In each interview, only questions applicable to that particular interviewee would be asked.

1. Please briefly introduce your company/organization and your personal experience with toll road development in the United States since the 1980s, with a particular focus on public-private partnership projects.
2. What is the most interesting case(s) you have been involved with which could provide significant information about how the public and private sectors negotiate to shape partnerships and reach agreements.
3. As a consultant working for a private consortium, when you review a Request for Proposals (RFP) from a government agency or consult on an unsolicited proposal to government, what factors or characteristics of the project would lead you to recommend submitting a proposal or not? In other words, what makes you think a project is worth doing?
   a. What about projects that look somewhat promising but also contain a high degree of risk? What types of risk are particularly troubling? How will this affect the proposal of the private consortium, if they still submit one?
   b. At what point will the bidding consortia have a general idea about the capital structure? What do you look for in the RFP to guide your recommendation about submitting a bid?
4. When preparing a proposal or consulting with government agencies, what would a private developer do in order to stand out among other developers and increase its chance of winning?
5. When a public agency reviews a proposal, what are they looking for and what factors would help them select a private partner?
   a. Besides access to private resources, what do you think really drives government to do partnerships? What are the critical factors, please give an example?
   b. Almost all states are facing transportation funding shortfalls, but not many of them are considering or doing public-private partnerships. Why do you think this is?
6. If you are part of the project development team, when you begin negotiations on project details, what items are on your negotiation list and in what order? What would be the typical debate(s) over each item? Especially for risks associated with the environmental review process, land acquisition, and revenue uncertainty, what would be the negotiating points for each one of them? How would a private developer propose in order to mitigate these risks?
7. I am looking for recommendations for interesting and informative cases which would illustrate significant characteristics of partnership formation and how agreement was reached on the nature of the partnership.
8. Overall, do you think public-private partnership agreements/projects have gotten better over time in terms of mutual benefit and reciprocity? How much better for the public sector, and how much better for the private sector? How did this occur and what are major things the parties have learned from the past experience?
9. There is some question about whether the capital structure determines the nature of partnership or the nature of

partnership determines capital structure arrangements. What is your opinion?
10. Permission for tape recorder.

Conflict of interest statement

I declare that I have no financial and personal relationships with other people or organizations that can inappropriately influence my work, there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of, the manuscript entitled, "evolution of public-private partnership models in American toll road development: learning based on public institutions' risk management".

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